1. (50 pts) An experiment was conducted in order to investigate four different treatments of pasture and two mineral supplements on milk yield. The total number of cows available was 24. The experiment was replicated in three different seasons of the year. The following milk yields were measured:

Block	Pasture	Mineral	Milk (kg)
1	4	2	30
1	4	1	29
1	1	2	27
1	1	1	25
1	2	1	26
1	2	2	28
1	3	2	26
1	3	1	24
2	2	1	32
2	2	2	37
2	1	2	30
2	1	1	31
2	4	1	34
2	4	2	37
2	3	1	33
2	3	2	32
3	1	2	34
3	1	1	31
3	2	1	30
3	2	2	3
3	4	2	36
3	4	1	38
3	3	1	33
3	3	2	32

a. State the hypotheses. What experimental design is this? (10 pts)
b. Determine which factor is hard to change and which one is not. (10 pts)
c. How many actual factors in this design, list them out. (10 pts)
d. Make conclusion on the effects of the treatment factors on the milk yield.

2. (50 pts) Below is the data obtained from a Box-Behnken experiment to minimize the output M of a process.

a. State the designing process to explain the design; your answer should include the purpose of the experiment, the process of choosing the levels and how you are going to use the data obtained from the experiment. (15 pts)
b. Perform appropriate calculations to determine the optimized parameters using the data in the table. (25 pts)
c. What is the last step you need to do in this optimization process? (10 pts)

Box-Behnken Experiment number	Parameters [mm]			Results
	L	D_{in}	D_{out}	M [g]
1	80	13	18	76.45
2	80	16	18	33.54
3	100	13	18	95.56
4	100	16	18	41.92
5	80	14.50	17	38.84
6	80	14.50	19	74.35
7	100	14.50	17	48.55
8	100	14.50	19	92.94
9	90	13	17	66.59
10	90	13	19	106.5
11	90	16	17	18.31
12	90	16	19	58.26
13	90	14.50	18	63.12